

Name: _____

St George Girls High School

Trial Higher School Certificate Examination

2014



Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen.
- Write your student number on each booklet.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- The mark allocated for each question is listed at the side of the question.
- Marks may be deducted for careless or poorly presented work.

Total Marks – 100

Section I – Pages 2 – 6

10 marks

- Attempt Questions 1 – 10.
- Allow about 15 minutes for this section.
- Answer on the sheet provided.

Section II – Pages 7 – 14

90 marks

- Attempt Questions 11 – 16.
- Allow about 2 hours 45 minutes for this section.
- Begin each question in a new booklet.
- Show all necessary working in Questions 11 – 16.
- Templates for Q13(a) to be detached and placed in Q13 answer booklet.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Section I

10 marks

Marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

-
1. Which expression is equal to $\int \cos^3 x \, dx$?
- (A) $\frac{1}{4} \sin^4 x + C$
- (B) $\sin x - \frac{\sin^3 x}{3} + C$
- (C) $\sin x + \frac{\cos^3 x}{3} + C$
- (D) $\cos x - \frac{\sin^3 x}{3} + C$
2. The eccentricity of the hyperbola with the equation $\frac{x^2}{3} - \frac{y^2}{4} = 1$ is:
- (A) $1 + \frac{2}{\sqrt{3}}$
- (B) $\sqrt{\frac{7}{3}}$
- (C) $\frac{\sqrt{21}}{3}$
- (D) $\frac{5}{3}$
3. Let the point R represent the complex number z on an Argand diagram. Which of the following describes the locus of R specified by $2|z| = z + \bar{z} + 4$.
- (A) Circle with centre $(0, 0)$ and radius 4
- (B) Parabola with vertex $(-1, 0)$, axis $y = 0$
- (C) Parabola with vertex $(-1, 0)$, axis $x = 0$
- (D) Perpendicular bisector of $(0, 0)$ and $(0, 4)$

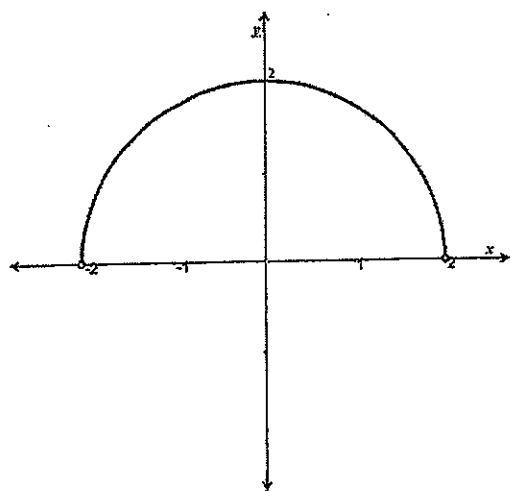
Section I (cont'd)

4. The polynomial $P(x) = x^4 + ax^3 - bx^2 - 12x$ has a double root at $x = -2$. What are the values of a and b .

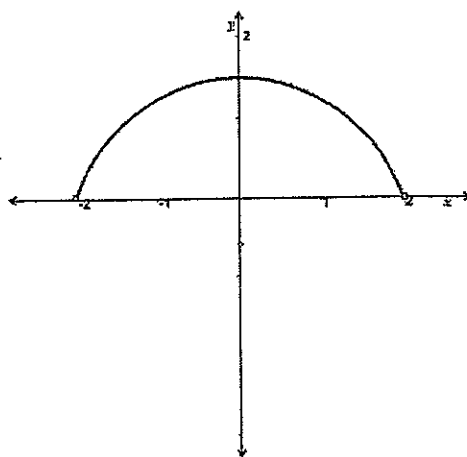
- (A) $a = -1$ and $b = -8$
 (B) $a = 8$ and $b = 1$
 (C) $a = 2$ and $b = 4$
 (D) $a = 1$ and $b = 8$

5. The locus of z if $\arg(z - 2) - \arg(z + 2) = \frac{\pi}{4}$ is best shown as:

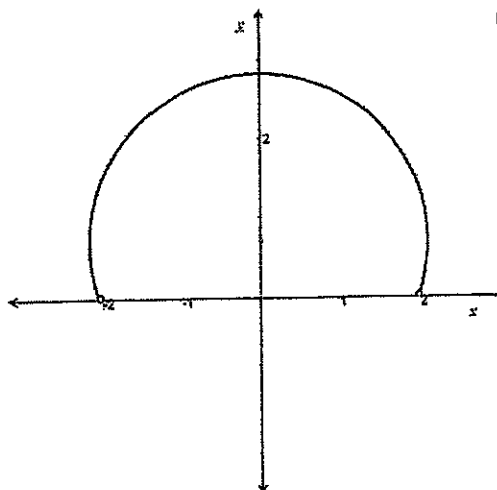
(A)



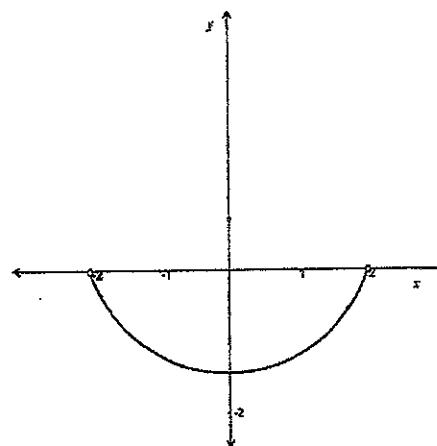
(B)



(C)



(D)



Section I (cont'd)

6. The derivative of the curve $x^3 + 9x^2 - y^2 + 27x - 4y + 23 = 0$ is:

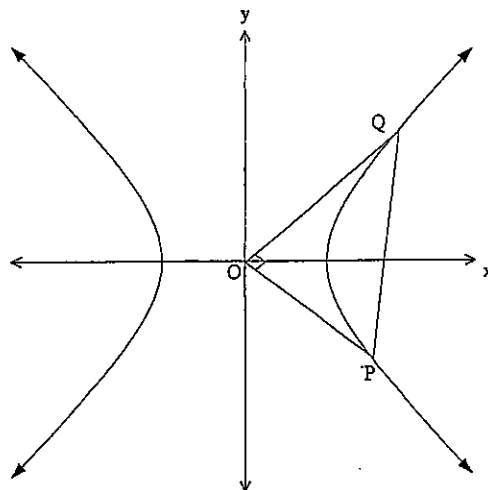
(A) $\frac{dy}{dx} = \frac{x^2+6x+9}{2y}$

(B) $\frac{dy}{dx} = \frac{3x^2+18x+27}{2y+4}$

(C) $\frac{dy}{dx} = \frac{3x^2+18x+27}{-(2y+4)}$

(D) $\frac{dy}{dx} = \frac{x^2+6x+9}{-2y}$

7. The diagram below shows the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where $a > b > 0$. The points $P(a \sec \theta, b \tan \theta)$ and $Q(a \sec \alpha, b \tan \alpha)$ lie on the hyperbola and the chord PQ subtends a right angle at the origin.



Use the parametric representation of the hyperbola to determine which of the following expressions is correct?

(A) $\sin \theta \sin \alpha = -\frac{a^2}{b^2}$

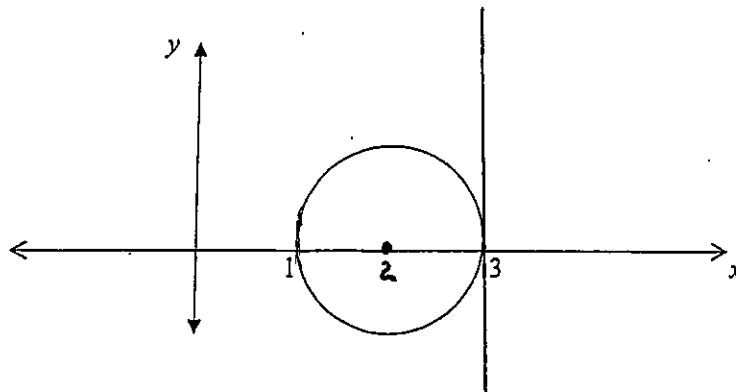
(B) $\sin \theta \sin \alpha = \frac{a^2}{b^2}$

(C) $\tan \theta \tan \alpha = -\frac{a^2}{b^2}$

(D) $\tan \theta \tan \alpha = \frac{a^2}{b^2}$

Section I (cont'd)

8.



In the diagram above the circle $(x - 2)^2 + y^2 = 1$ is shown. The region bounded by the circle is rotated about the line $x = 3$. Using the method of cylindrical shells the volume of the solid of revolution so formed is given by:

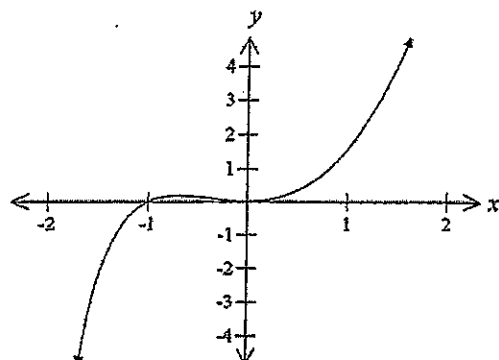
- (A) $V = 4\pi \int_1^3 (x - 3)\sqrt{1 - (x - 2)^2} dx$
- (B) $V = 4\pi \int_2^3 (3 - x)\sqrt{1 - (x - 2)^2} dx$
- (C) $V = 2\pi \int_1^3 (3 - x)\sqrt{1 - (x - 2)^2} dx$
- (D) $V = 2\pi \int_1^3 (3 - x)\sqrt{1 - (x - 2)^2} dx$

9. Let $P(x)$ be a polynomial of degree $n > 0$ such that $P(x) = (x - \alpha)^p \cdot Q(x)$, where $p \geq 2$ and α is a real number. $Q(x)$ is a polynomial with real coefficients of degree $q > 0$. Which of the following is definitely an incorrect statement?

- (A) $P(x)$ changes sign around the root $x = \alpha$
- (B) $n \leq p + q$
- (C) Roots of $Q(x)$ are conjugates of one another
- (D) $P'(\alpha) > 0$

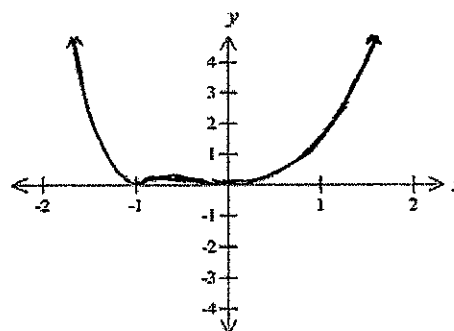
Section I (cont'd)

10. The diagram shows the graph of the function $y = f(x)$.

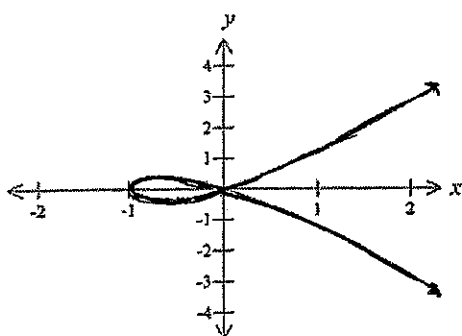


The diagram that shows the graph of the function $y = [f(x)]^2$ is:

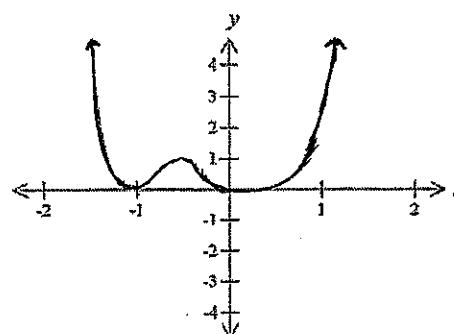
(B)



(C)



(D)



End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

- | Question 11 (15 marks) Use a SEPARATE writing booklet | Marks |
|---|-------|
| a) Given $z = \frac{1-i}{\sqrt{3}+i}$. | |
| (i) Find the modulus $ z $ and argument $\arg(z)$ of z . | 2 |
| (ii) Find the smallest positive integer n such that z^n is REAL. | 2 |
| b) Find the complex square roots of $1 - 2\sqrt{2}i$ giving your answers in the form $x + iy$ where x and y are real. | 3 |
| c) Find the three different values of z for which $z^3 = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$. | 3 |
| d) (i) On an Argand Diagram, draw and shade the region R given by
$ z - 3 - 3i \leq 3$. | 1 |
| (ii) P is a point in the region R , representing the complex number z .
What is the maximum value of $ z $? | 2 |
| (iii) The tangent to the curve at P cuts the x -axis at the point T . By using the nature of $\triangle OPT$, or otherwise, find the exact area of $\triangle OPT$. | 2 |

Question 12 (15 marks) Use a SEPARATE writing booklet

Marks

- a) Using the substitution $t = \tan \frac{x}{2}$, evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{5 + 3 \cos x}$. 4

[leave your answer in EXACT form]

- b) Find $\int \sin^4 x \cos^3 x \, dx$. 3

- c) (i) Show that $(1 + t^2)^{n-1} + t^2 (1 + t^2)^{n-1} = (1 + t^2)^n$. 1

- (ii) Let $I_n = \int_0^x (1 + t^2)^n \, dt$ for n a positive integer. 4

Use integration by parts, and part (i) above, to show that

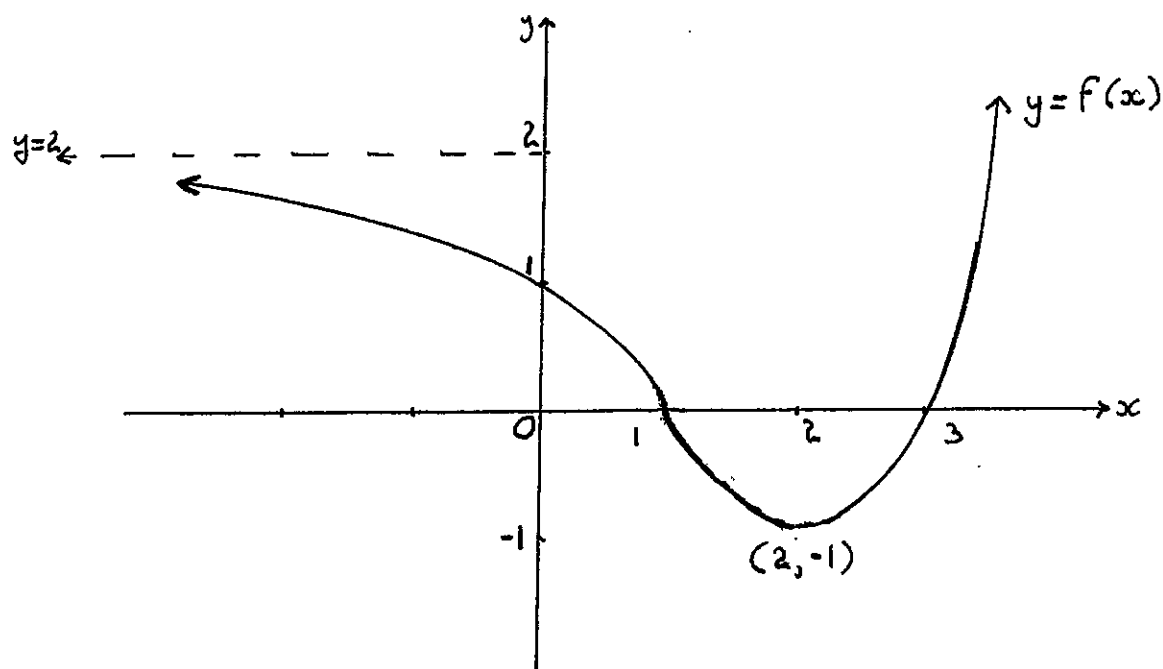
$$I_n = \frac{1}{2n+1} x(1+x^2)^n + \frac{2n}{2n+1} I_{n-1}.$$

- d) Make a suitable substitution to find the exact value of $\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{x} \cdot \sqrt{1-x}}$. 3

Question 13 (15 marks) Use a SEPARATE writing booklet

Marks

- a) The diagram below shows the graph of a function $f(x)$.



Using the separate templates of the graph of $y = f(x)$ provided at the end of this paper, sketch the graphs of:

- | | |
|----------------------|---|
| (i) $y = [f(x)]^2$. | 2 |
| (ii) $y = f'(x)$. | 2 |
| (iii) $y^2 = f(x)$. | 2 |

Question 13 (cont'd)

Marks

- b) (i) Show that the equation of the normal to the hyperbola $xy = c^2$ at $P\left(cp, \frac{c}{p}\right)$ is $p^3x - py = c(p^4 - 1)$.

2

- (ii) The normal at $P\left(cp, \frac{c}{p}\right)$ meets the hyperbola $xy = c^2$ again at $Q\left(cq, \frac{c}{q}\right)$.

2

Prove that $p^3q = -1$.

- (iii) Hence, show that if $M(x, y)$ is the midpoint of PQ , then $\frac{x}{y} = -\frac{1}{p^2}$.

3

- c) Sketch on an Argand Diagram (at least $\frac{1}{3}$ of a page) the locus of the complex number z where $\arg(z + 1) = \arg(z - 1 + i)$.

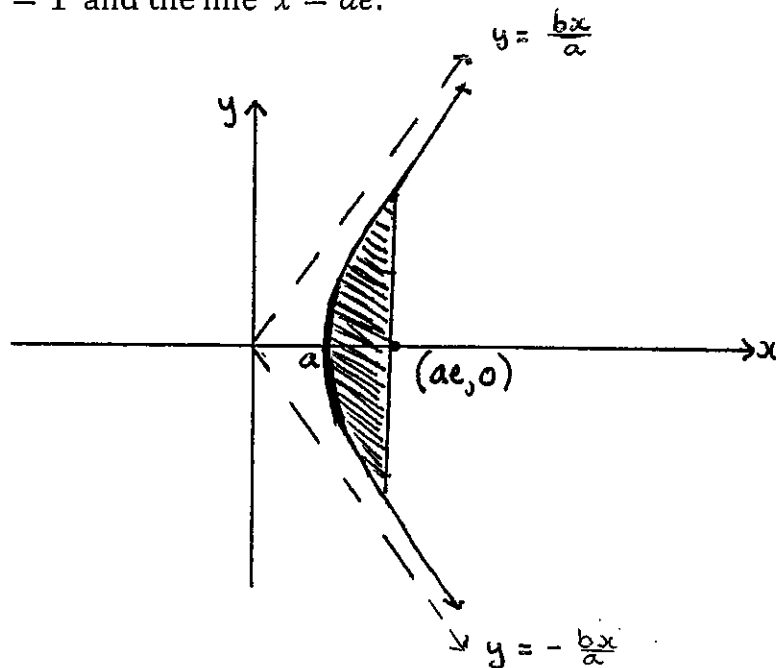
2

Question 14 (15 marks) Use a SEPARATE writing booklet

Marks

- a) The shaded region shown below represents the area bounded by the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and the line $x = ae$.

4



Find the volume generated by rotating this area about the y -axis through 360° (answer in terms of a, b).

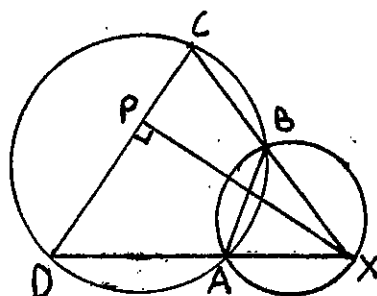
- b) When a polynomial $P(x)$ is divided by $(x - 3)$ the remainder is -2 . When $P(x)$ is divided by $(x + 2)$ the remainder is 8 . Find the remainder when $P(x)$ is divided by $(x - 3)(x + 2)$.

3

Question 14 (cont'd)

Marks

c)



In the diagram above, $AB = AD = AX$ and $XP \perp DC$.

(i) Prove that $\angle DBX = 90^\circ$.

2

(ii) Hence, or otherwise, prove that $\angle APB = \angle ABP$.

2

d) The three non-zero roots of the equation $x^3 - 3px + q = 0$ are α, β, γ .

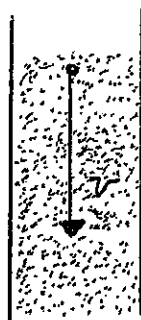
4

Find the monic equation whose roots are $\frac{\beta\gamma}{\alpha}, \frac{\alpha\gamma}{\beta}, \frac{\alpha\beta}{\gamma}$ expressing its coefficients in terms of p and q .

Question 15 (15 marks) Use a SEPARATE writing booklet

Marks

- a) Consider a particle falling through a fluid as shown in the diagram below:



The resistive frictional force on the particle is proportional to its velocity. That is, the resistance force may be written as $R = -mkv$ where k is a constant and the particles velocity is $v(ms^{-1})$.

- (i) If the particle falls vertically from rest, show that the terminal velocity v_T is given by $v_T = \frac{g}{k}$, where $g(ms^{-2})$ is the acceleration due to gravity. 3

- (ii) If the particle is projected upwards into the resistive fluid with speed v_T , show that after t seconds.

(α) its speed $v(ms^{-1})$ is given by $v = v_T(2e^{-kt} - 1)$. 3

(β) its height, $x(m)$ is given by $x = \frac{v_T}{k}(2 - kt - 2e^{-kt})$. 3

- (iii) Hence, show that the greatest height that the particle can reach is 2

$$x_{\max} = \frac{v_T}{k}(1 - \ln 2).$$

- b) The equation $(\sin^2 \theta) z^2 - (\sin 2\theta) z + 1 = 0$, where $0 < \theta < \frac{\pi}{2}$, has roots α and β .

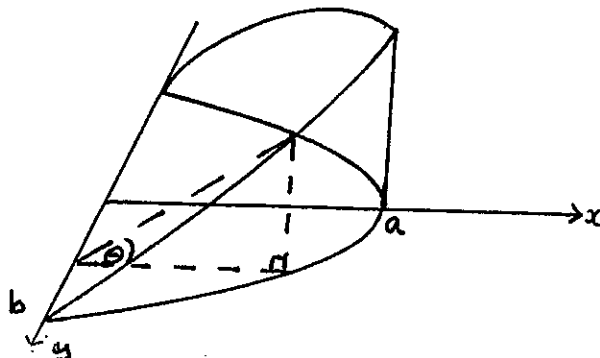
- (i) Show that the roots of the equation are $(\cot \theta + i)$ and $(\cot \theta - i)$. 2

- (ii) Hence, show that $\alpha^n + \beta^n = \frac{2 \cos n \theta}{\sin^n \theta}$. 2

Question 16 (15 marks) Use a SEPARATE writing booklet

Marks

- a) A solid in the shape of a wedge has its base in half an ellipse, with major axis $2a$ and minor axis $2b$. Cross-sections taken perpendicular to the base are all right angled triangles and the angle between the two flat surfaces of the wedge is θ° .



- (i) Show that the area of the triangular face of the cross-section is given by

3

$$\frac{a^2}{2b^2} (b^2 - y^2) \tan \theta.$$

- (ii) Hence, or otherwise, find the volume of the wedge giving your answer in terms of a , b and $\tan \theta$.

3

- b) (i) Prove that

$$(\alpha) \quad \frac{{}^1C_0}{x} - \frac{{}^1C_1}{x+1} = \frac{1!}{x(x+1)}.$$

2

$$(\beta) \quad \frac{{}^2C_0}{x} - \frac{{}^2C_1}{x+1} + \frac{{}^2C_2}{x+2} = \frac{2!}{x(x+1)(x+2)}.$$

- (ii) Given $T(k, x) = \frac{k!}{x(x+1)(x+2)\dots(x+k)}$, prove that

3

$$T(k, x) - T(k, x+1) = T(k+1, x).$$

- (iii) Hence prove, using Mathematical Induction or otherwise, that for $n \geq 1$:

4

$$\frac{{}^nC_0}{x} - \frac{{}^nC_1}{x+1} + \frac{{}^nC_2}{x+2} - \frac{{}^nC_3}{x+3} + \dots + (-1)^n \frac{{}^nC_n}{x+n} = \frac{n!}{x(x+1)(x+2)(x+3)\dots(x+n)}.$$

[You may use the result: ${}^{k+1}C_r = {}^kC_r + {}^kC_{r-1}$]

[Note: ${}^{k+1}C_0 = {}^kC_0$ and ${}^{k+1}C_{k+1} = {}^kC_k$]

End of Paper

Student Number: _____ Class Teacher: _____

Section I

Year 12 Trial HSC Examination 2014

Mathematics Extension 2

Multiple-choice Answer Sheet - Questions 1 – 10

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A  B  C  D 

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A  B  C  D 

- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

Staff Use Only

Section I	/10
Section II	/90
Question 11	/15
Question 12	/15
Question 13	/15
Question 14	/15
Question 15	/15
Question 16	/15
Total	/100

EXT 2 SOLUTIONS TRIAL 2014

Section I

$$\begin{aligned}
 \text{Q1 } \int \cos x (1 - \sin^2 x) dx \\
 &= \int [\cos x - \cos x \cdot \sin^2 x] dx \quad \textcircled{B} \\
 &= \sin x - \frac{1}{3} \sin^3 x + C
 \end{aligned}$$

$$\begin{aligned}
 \text{Q2 } e^2 &= 1 + \frac{b^2}{a^2} \\
 e &= \sqrt{1 + \frac{4}{3}} \quad \textcircled{B} \\
 &= \sqrt{\frac{7}{3}} = \frac{\sqrt{21}}{3} \quad \text{or } \textcircled{C}
 \end{aligned}$$

$$\begin{aligned}
 \text{Q3 } 2\sqrt{x^2 + y^2} &= 2x + 4 \\
 x^2 + y^2 &= x^2 + 4x + 4 \quad \textcircled{B} \\
 y^2 &= 4(x + 1)
 \end{aligned}$$

$$\begin{aligned}
 \text{Q4 } P'(x) &= 4x^3 + 3ax^2 - 2bx - 12 \\
 P'(2) &= -32 + 12a + 4b - 12 \\
 \therefore 12a + 4b &= 44 \quad \textcircled{D} \\
 3a + b &= 11 \quad \text{--- (I)} \\
 P(-2) &= 16 - 8a - 4b + 24 \\
 \therefore 8a + 4b &= 40 \\
 2a + b &= 10 \quad \text{--- (II)} \\
 \text{(I)} - \text{(II)} \quad a &= 1 \quad \therefore b = 8
 \end{aligned}$$

Q5.

(C)

Q6. $3x^2 + 18x - 2y \cdot \frac{dy}{dx} + 27 - 4 \cdot \frac{dy}{dx} = 0$

$$\therefore (2y + 4) \frac{dy}{dx} = 3x^2 + 18x + 27$$

$$\frac{dy}{dx} = \frac{3x^2 + 18x + 27}{2y + 4}$$

(B)

Q7 $m_1 = \frac{b \tan \alpha}{a \sec \alpha}$ & $m_2 = \frac{b \tan \theta}{a \sec \theta}$

$b \Rightarrow m_1 \times m_2 = -1$

$$\therefore \frac{b^2 \tan \alpha \tan \theta}{a^2 \sec \alpha \sec \theta} = -1$$

$$\frac{b^2}{a^2} = - \frac{\sec \alpha \sec \theta}{\tan \alpha \tan \theta}$$

$$- \frac{b^2}{a^2} = \frac{1}{\frac{\cos \alpha \cos \theta}{\sin \alpha \sin \theta}}$$

(A)

$$- \frac{b^2}{a^2} = \frac{1}{\sin \alpha \sin \theta}$$

Q8 $\delta V = (3-x) \cdot 2\pi \cdot 2\sqrt{1-(x-2)^2} dx$

(D)

$$\therefore V = 4\pi \int_1^3 (3-x) \sqrt{1-(x-2)^2} dx$$

Q9

(D)

Q10.

(B)

Section 2

Question Number: 11

$$(a)(i) \quad |z| = \frac{|1-i|}{|\sqrt{3}+i|}$$

$$\text{or } z = \frac{1-i}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i}$$

$$= \frac{\sqrt{1^2 + (-1)^2}}{\sqrt{(\sqrt{3})^2 + 1^2}}$$

$$= \frac{(\sqrt{3}-1) - i(\sqrt{3}+1)}{4}$$

$$= \frac{\sqrt{2}}{2}$$

$$|z| = \sqrt{\left[\frac{(\sqrt{3}-1)}{4}\right]^2 + \left[\frac{(\sqrt{3}+1)}{4}\right]^2}$$

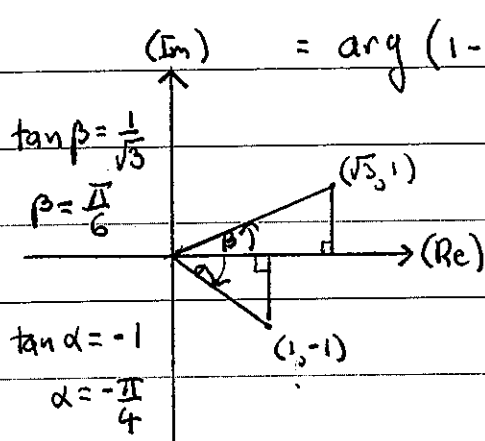
$$= \frac{1}{4} \sqrt{4 - 2\sqrt{3} + 4 + 2\sqrt{3}}$$

$$= \frac{\sqrt{2}}{2}$$

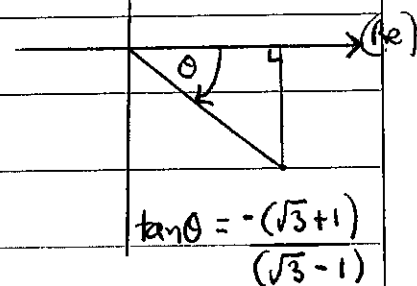
and

$$\arg(z) = \arg\left[\frac{(1-i)}{(\sqrt{3}+i)}\right]$$

$$\text{or } z = \frac{(\sqrt{3}-1)}{4} - \frac{(\sqrt{3}+1)}{4}i$$



$$\begin{aligned} \arg(z) &= \arg(1-i) - \arg(\sqrt{3}+i) \\ &= \left(-\frac{\pi}{4}\right) - \left(\frac{\pi}{6}\right) \\ &= -\frac{5\pi}{12} \end{aligned}$$



$$\begin{aligned} \tan \theta &= \frac{-(\sqrt{3}+1)}{(\sqrt{3}-1)} \\ &= -(2+\sqrt{3}) \\ \therefore \theta &= -\frac{5\pi}{12} \end{aligned}$$

$$(ii) \quad z = \frac{\sqrt{2}}{2} \operatorname{cis} \left(-\frac{5\pi}{12}\right)$$

$$\text{Then } z^n = \left(\frac{\sqrt{2}}{2}\right)^n \operatorname{cis} \left(-\frac{5\pi n}{12}\right)$$

De Moivre's
Theorem

$$= \left(\frac{\sqrt{2}}{2}\right)^n \left(\cos \left(-\frac{5\pi n}{12}\right) + i \sin \left(-\frac{5\pi n}{12}\right)\right)$$

$$= \left(\frac{\sqrt{2}}{2}\right)^n \left(\cos \left(\frac{5\pi n}{12}\right) - i \sin \left(\frac{5\pi n}{12}\right)\right)$$

Require $\sin\left(\frac{5n\pi}{12}\right) = 0$ if z^n is REAL

$$\therefore \frac{5n\pi}{12} = 0, \pi, 2\pi, \dots$$

$$\frac{5n}{12} = 0, 1, 2, \dots \quad \text{Req } n \text{ to be a multiple of 12}$$

$$\begin{aligned} \text{When } n=12, \quad z^{12} &= (2^{-\frac{1}{2}})^{12} (\cos(5\pi) - i \sin(5\pi)) \\ &= 2^{-6} \cdot 1 \\ &= -\frac{1}{64} \end{aligned}$$

(b) let $(x+iy)^2 = 1 - i \cdot 2\sqrt{2}$

then $x^2 - y^2 + i \cdot 2xy = 1 - i \cdot 2\sqrt{2}$

So equating Real and Imaginary parts

$$x^2 - y^2 = 1 \dots (I)$$

$$2xy = -2\sqrt{2} \dots (II) \Rightarrow \frac{x^2 - 2}{x^2} = 1$$

$$y = -\frac{\sqrt{2}}{x} \quad x^2 - x^2 - 2 = 0$$

$$\text{gives } x^2 = \frac{1 \pm \sqrt{9}}{2}$$

Since x is REAL $x^2 = 2$

$$x = \pm\sqrt{2}$$

Square root $(\sqrt{2} - i)$ and $(-\sqrt{2} + i)$

(c) z is cube root of $\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}} = w$

$$|w| = 1$$

$$\therefore w = \cos\left(\frac{\pi}{4} + 2k\pi\right) + i \sin\left(\frac{\pi}{4} + 2k\pi\right)$$

$$\arg(z) = \frac{\pi}{4}$$

Let $z = r(\cos \theta + i \sin \theta)$

$|z| = |w|$
 $= 1$

then $z^3 = (\cos 3\theta + i \sin 3\theta)$ by de Moivre's

Equating Real & imaginary parts

$\cos(3\theta) = \cos\left(\frac{\pi}{4} + 2k\pi\right) \quad \sin 3\theta = \sin\left(\frac{\pi}{4} + 2k\pi\right)$

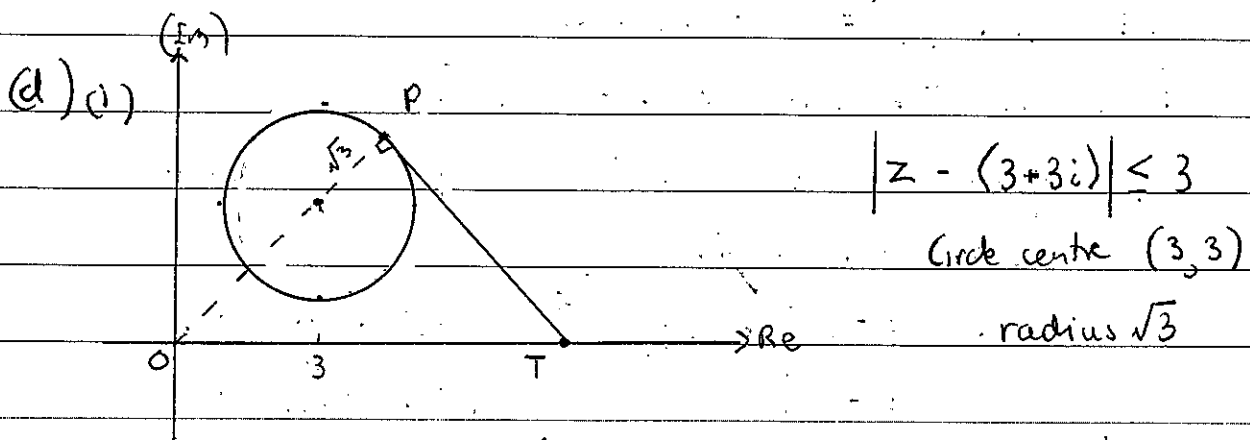
$3\theta = \frac{\pi}{4} + 2k\pi$

$\therefore \theta = \frac{\pi}{12}(1+8k)$

When $k=0$, $z_1 = \cos\left(\frac{\pi}{12}\right) + i \sin\left(\frac{\pi}{12}\right)$

$k=1$, $z_2 = \cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) = -\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right)$

$k=2$, $z_3 = \cos\left(\frac{17\pi}{12}\right) + i \sin\left(\frac{17\pi}{12}\right) = -\cos\left(\frac{5\pi}{12}\right) - i \sin\left(\frac{5\pi}{12}\right)$



(ii) Max. value $|z|$ is $(|3+3i| + \text{radius of circle})$
 $= \sqrt{3^2+3^2} + \sqrt{3}$
 $= 3\sqrt{2} + \sqrt{3}$

(iii) $\arg(z) = \arg(3+3i)$
 $= \frac{\pi}{4}$

then $\triangle OPT$ is isosceles right angle $\triangle$

gives $OP = OT$

$\therefore \text{Area } \triangle = \frac{1}{2} (OP \cdot OT)$
 $= \frac{1}{2} (3\sqrt{2} + \sqrt{3})^2$



Start here for

Question Number: 12

(a) Let $t = \tan \frac{x}{2}$

then $x = 2 \tan^{-1} t$

$$\frac{dx}{dt} = \frac{2}{1+t^2}$$

On Substitution

$$\int_0^1 \frac{\frac{2}{1+t^2} dt}{5 + 3 \frac{(1-t^2)}{1+t^2}}$$

When $x=0$, $t=0$

$x=\frac{\pi}{2}$, $t=1$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$= \int_0^1 \frac{2 dt}{5(1+t^2) + 3(1-t^2)}$$

$$= \int_0^1 \frac{2 dt}{8 + 2t^2}$$

$$= \int_0^1 \frac{dt}{4 + t^2}$$

$$= \frac{1}{2} \left[\tan^{-1} \left(\frac{t}{2} \right) \right]_0^1$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \right)$$

(b) $\int \sin^4 x \cdot \cos^3 x dx = \int \sin^4 x \cdot (\cos^2 x) \cdot \cos x dx$

$$= \int \sin^4 x (1 - \sin^2 x) \cdot \cos x dx$$

$$= \int \sin^4 x \cdot \cos x dx - \int \sin^6 x \cdot \cos x dx$$

$$\left[\int f'(x) [f(x)]^n dx \right]$$

$$= \frac{1}{5} \sin^5 x - \frac{1}{7} \sin^7 x + C$$

(c)(i) L.H.S. = $(1+t^2)^{n-1} [1+t^2]$

$$= (1+t^2)^n$$

$$= \text{R.H.S.}$$

$$(ii) \text{ Let } I_n = \int_0^x (1+t^2)^n \cdot \frac{d}{dt}(t) \cdot dt$$

$$= [t \cdot (1+t^2)^n]_0^x - \int_0^x t \cdot n \cdot (1+t^2)^{n-1} \cdot 2t \, dt$$

$$= x \cdot (1+x^2)^n - 2n \int_0^x t^2 (1+t^2)^{n-1} \, dt$$

$$\text{Using part (i)} = x(1+x^2)^n - 2n \int_0^x [(1+t^2)^n - (1+t^2)^{n-1}] \, dt$$

$$I_n = x(1+x^2)^n - 2n I_n + 2n I_{n-1}$$

$$(1+2n) I_n = x(1+x^2)^n + 2n I_{n-1}$$

$$\therefore I_n = \frac{x \cdot (1+x^2)^n}{(1+2n)} + \frac{2n}{(1+2n)} \cdot I_{n-1}$$

$$(d) \text{ Let } u^2 = 1-x$$

$$\text{then } x = 1-u^2$$

$$\text{and } 2u \cdot du = -dx$$

$$\text{Substitution gives}$$

$$\int_1^{\frac{1}{\sqrt{2}}} \frac{-2u \cdot du}{\sqrt{1-u^2} \cdot u}$$

$$\text{When } x=0, u=1$$

$$x=\frac{1}{2}, u=\frac{1}{\sqrt{2}}$$

$$x>0, u>0$$

$$= 2 \int_{\frac{1}{\sqrt{2}}}^1 \frac{du}{\sqrt{1-u^2}}$$

$$= 2 \left[\sin^{-1} u \right]_{\frac{1}{\sqrt{2}}}^1$$

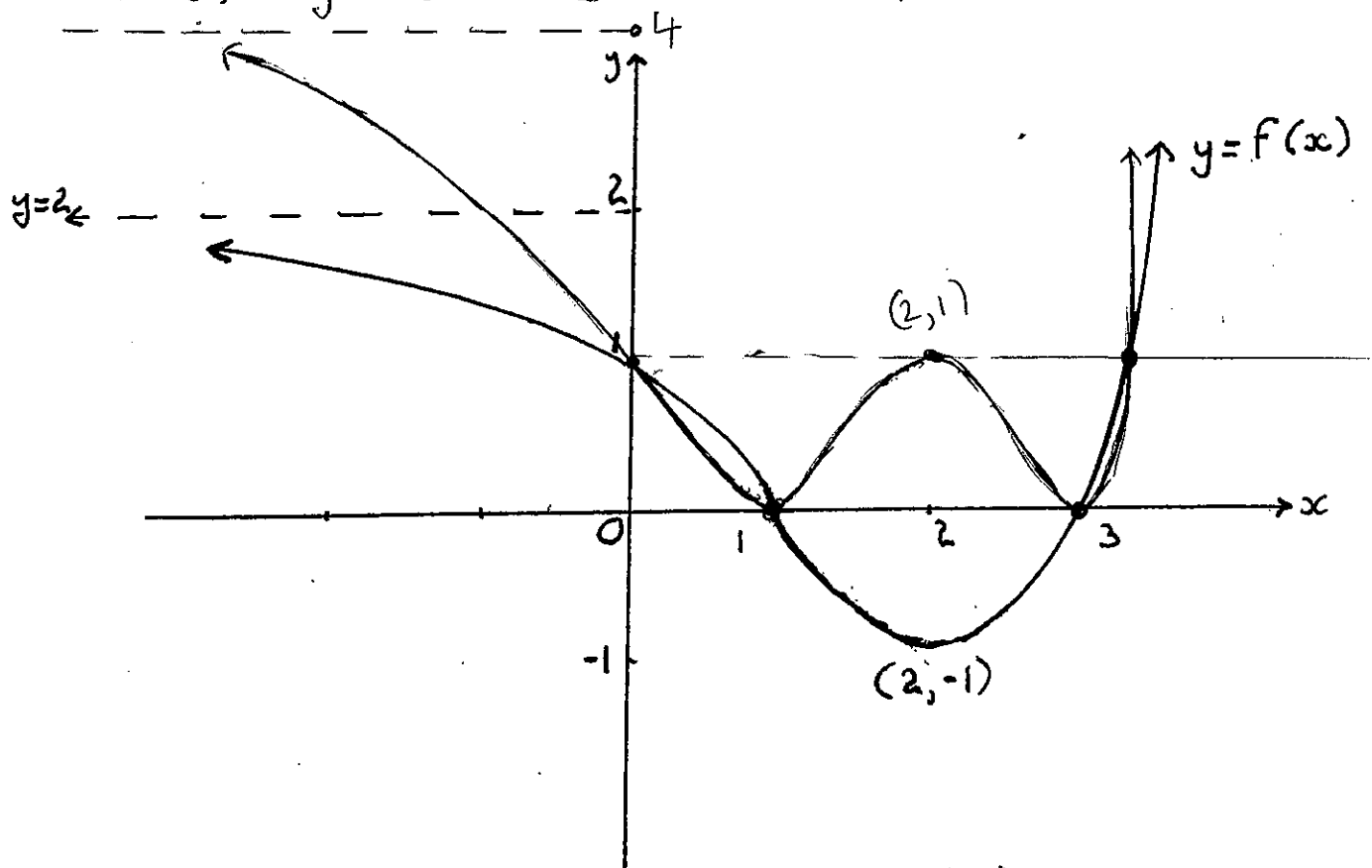
$$= 2 \left[\sin^{-1} 1 - \sin^{-1} \frac{1}{\sqrt{2}} \right]$$

$$= 2 \left[\frac{\pi}{2} - \frac{\pi}{4} \right]$$

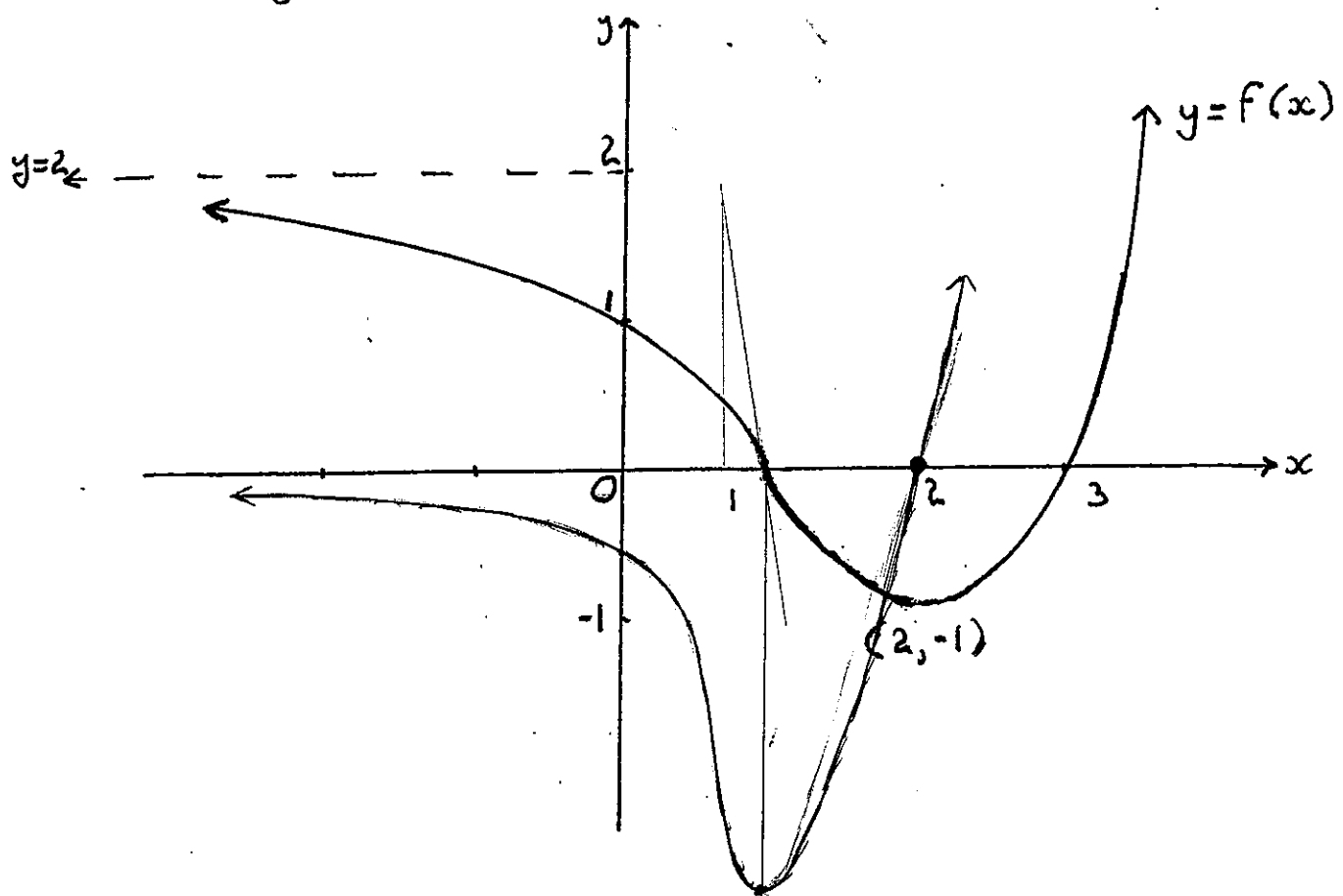
$$= \frac{\pi}{2}$$

QUESTION 13.

(a) (i) $y = [f(x)]^2$ (template)

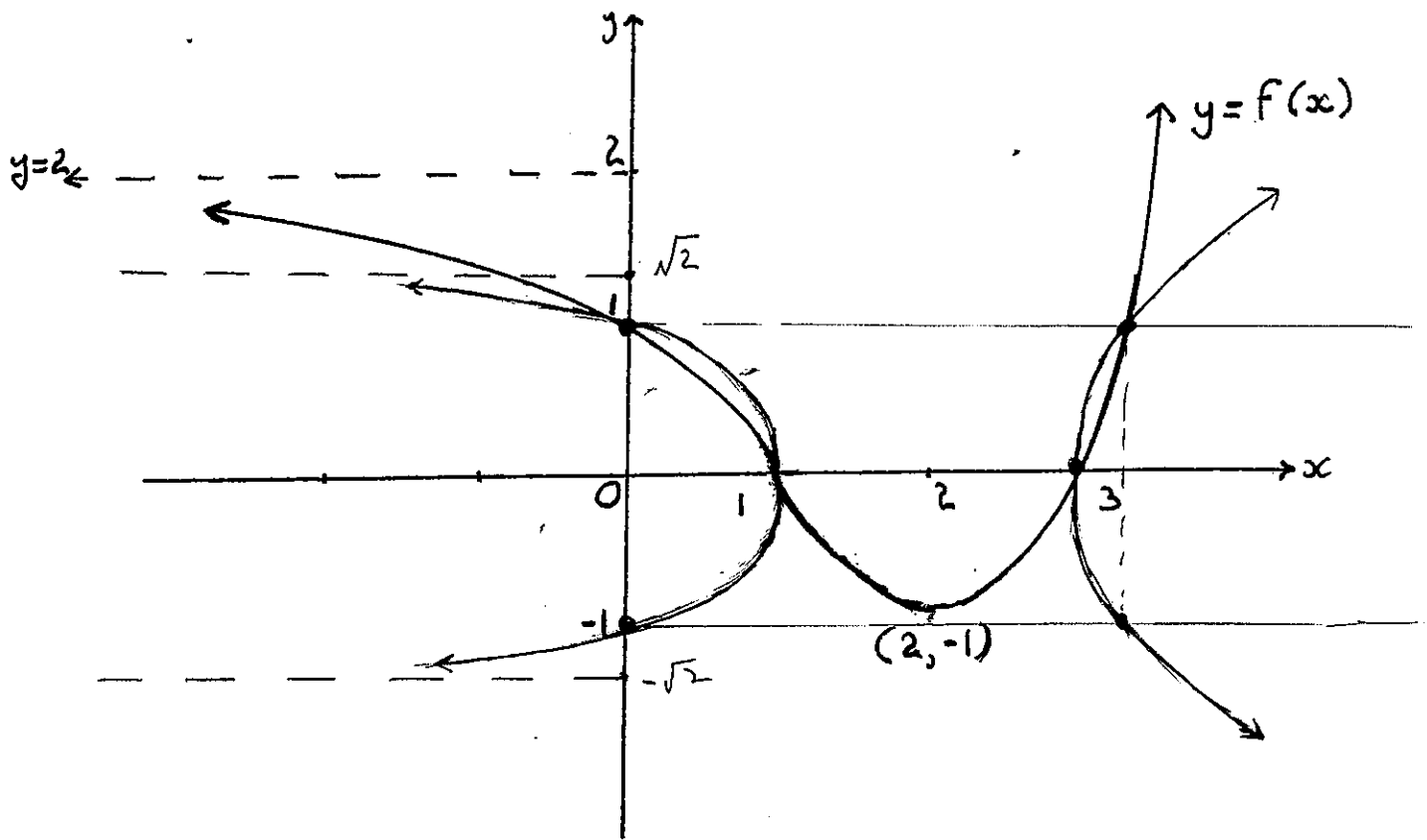


(a) (ii) $y = f'(x)$ (template)

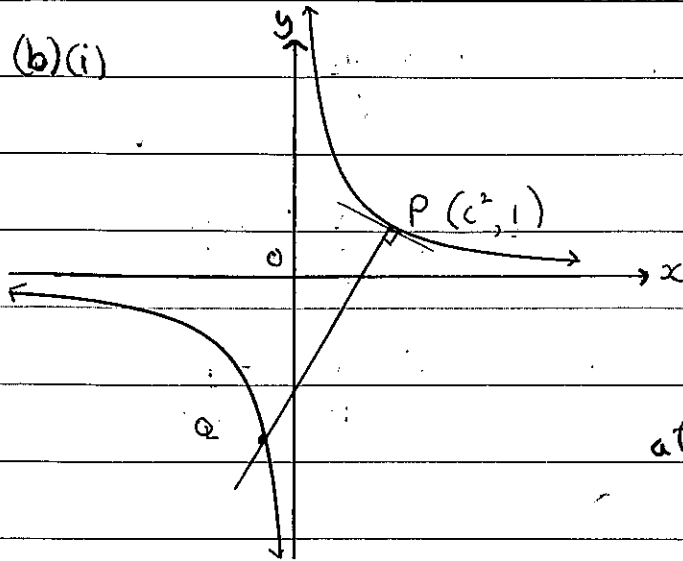


QUESTION 13.

(a) (iii) $y^2 = f(x)$ (template)



(b)(i)



$$xy = c^2 \quad \text{or} \quad y = \frac{c^2}{x}$$

$$\text{then } y + x \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

$$\text{at } P(cp, \frac{c}{p}), m_{\text{tangent}} = -\frac{\frac{c}{p}}{\frac{c}{p}} = -\frac{1}{p^2}$$

$\therefore$ Equation of tangent at P gradient p^2

$$y - \frac{c}{p} = p^2 (x - cp)$$

$$py - c = p^3 x - cp^4$$

$$p^3 x - py = c(p^4 - 1)$$

(ii) Q lies on hyperbola, let $Q(cq, \frac{c}{q})$ lie on normal through P, then

$$p^3 \cdot cq - p \cdot \frac{c}{q} = c(p^4 - 1)$$

$$p^3 q^2 - p = p^4 q - q$$

$$p^3 q (p - q) = - (p - q)$$

$$p^3 q = -1$$

$$(iii) M(x, y) \Rightarrow M\left(\frac{c(p+q)}{2}, \frac{\frac{c}{p}}{2} + \frac{\frac{c}{q}}{2}\right)$$

$$= M\left(\frac{c}{2}(p+q), \frac{c}{2} \cdot \frac{(p+q)}{pq}\right)$$

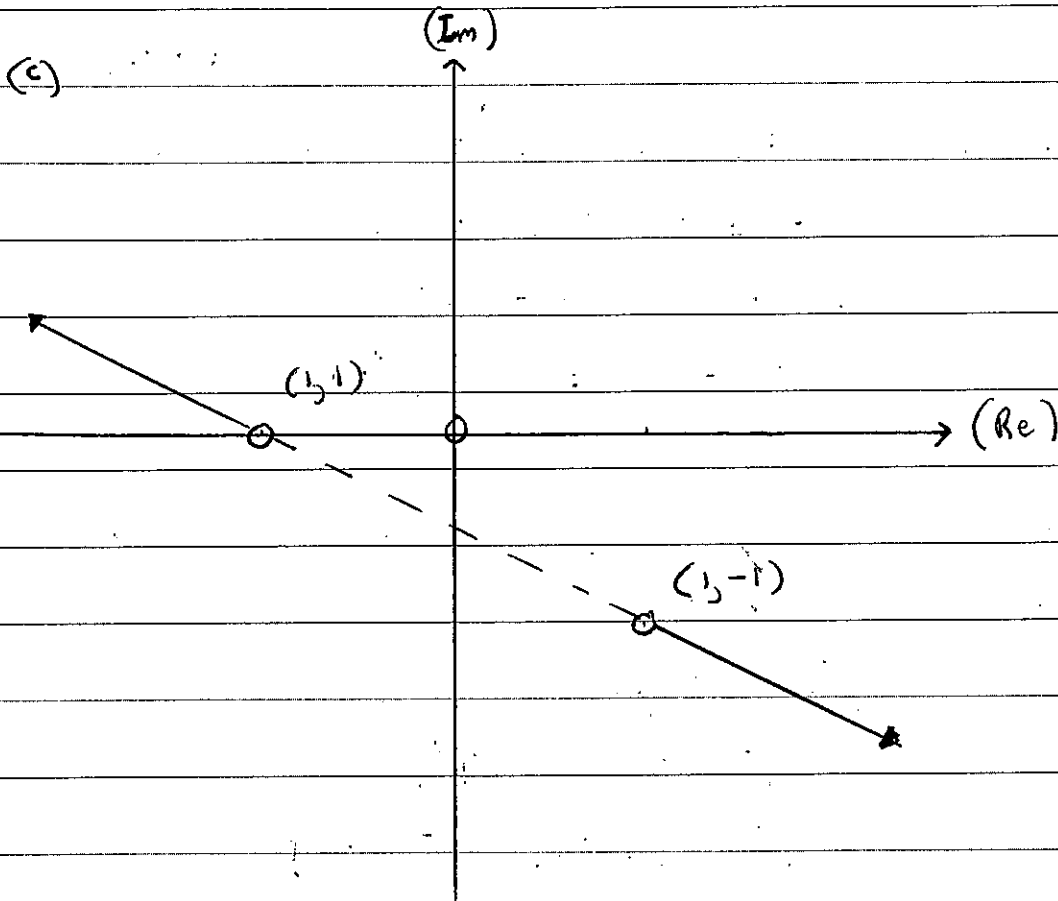
$$\text{Then, } x = \frac{c(p+q)}{2} \quad \text{--- (I) and } y = \frac{c(p+q)}{2pq} \quad \text{--- (II)}$$

Substitute (I) into II then $y = \frac{x}{pq}$

Now $p^3 q = -1 \Rightarrow \frac{x}{y} = pq$

$pq = -\frac{1}{p^2}$

$\therefore \frac{x}{y} = -\frac{1}{p^2}$

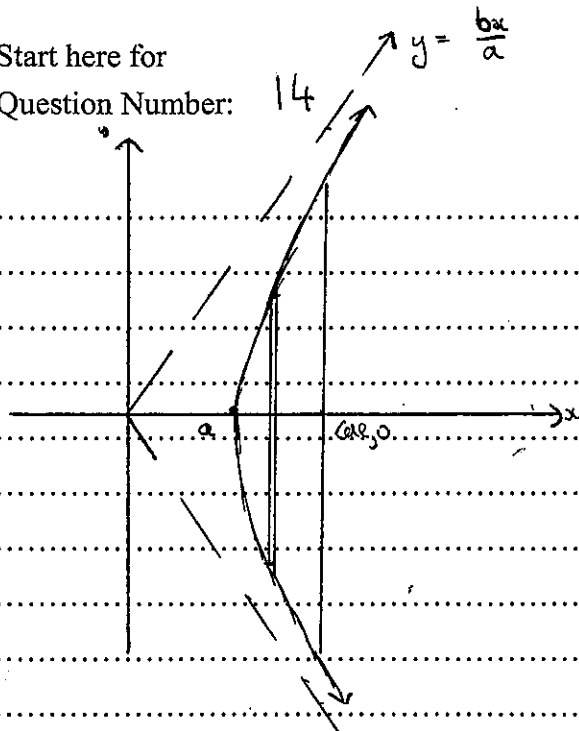


$\arg(z - (-1)) = \arg(z - (1 - i))$

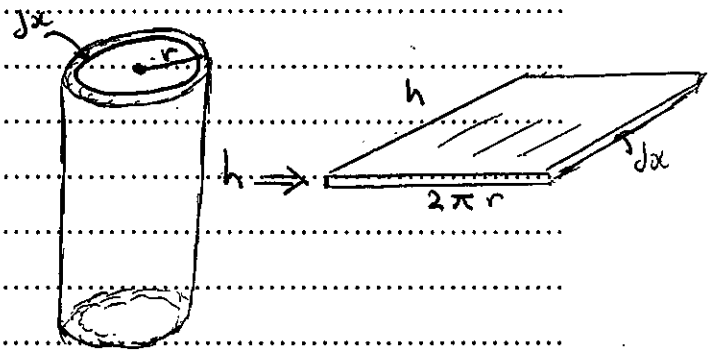


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Question Number: 14



by cylindrical shells



Elemental shell $\delta V = 2\pi r \cdot h \cdot \delta x$

Now $r = x$
and $h = 2y$ } $\therefore \delta V = 4\pi xy \cdot \delta x$

Given $\frac{y^2}{b^2} = \frac{x^2}{a^2} - 1$

$\therefore y = \pm \frac{b}{a} \sqrt{x^2 - a^2}$

Then $\delta V = 4\pi x \cdot \frac{b}{a} \sqrt{x^2 - a^2} \delta x$

$V = \lim_{\delta x \rightarrow 0} \sum_{x=a}^{ae} 4\pi x \cdot \frac{b}{a} \sqrt{x^2 - a^2} \cdot \delta x$

$= \frac{4\pi b}{a} \int_a^{ae} \frac{1}{2} \cdot 2x (x^2 - a^2)^{\frac{1}{2}} dx$

$= \frac{2\pi b}{a} \left[\frac{(x^2 - a^2)^{\frac{3}{2}}}{\frac{3}{2}} \right]_a^{ae}$

$= \frac{4\pi b}{3a} \left[(a^2 e^2 - a^2)^{\frac{3}{2}} - (a^2 - a^2)^{\frac{3}{2}} \right]$

$= \frac{4\pi b}{3a} \cdot a^3 (e^2 - 1)^{\frac{3}{2}}$

* $\frac{b^2}{a^2} = e^2 - 1$

$= \frac{4\pi a^2 b}{3} \left(\frac{b^2}{a^2} \right)^{\frac{3}{2}}$

$$\text{So } V = \frac{4\pi a^2 b}{3} \cdot \frac{b^3}{a^3}$$

$$= \frac{4\pi b^4}{3a}$$

(b) $P(x) = (x-3) \cdot Q(x) - 2$

and $P(x) = (x+2) \cdot H(x) + 8$

Now $P(x) = (x-3) \cdot B(x) + (ax+b)$

(I) $P(3) = 0 + 3a + b = -2$

$\therefore 3a + b = -2$

(II) $P(-2) = 0 - 2a + b = 8$

$\therefore -2a + b = 8$

So (I) - (II) gives $5a = -10$

$a = -2$

on substitution $b = 4$

Remainder is $(-2x + 4)$

(c) (i) DPX lie on a circle since DX subtends right angle at P . [angle in a semi-circle]

Since $AD = AX$ and DX is diameter, A is centre of circle and radius of circle $AD = AB$, so B lies on circle DPX .

DX diameter then subtends right angle at circumference, i.e. $\angle DBX = 90^\circ$.

(ii) Now $DPBX$ lie on circle with diameter DX , centre A .

So $AP = AB$ [radii of circle]

Gives $\triangle APB$ is isosceles.

then $\angle APB = \angle ABP$ (angles opposite equal sides are equal)

$$(d) (I) \alpha\beta\gamma = -q$$

$$\therefore \frac{\beta\gamma}{\alpha} = -\frac{q}{\alpha^2}$$

$$\frac{\alpha\beta}{\gamma} = -\frac{q}{\gamma^2}$$

$$\frac{\alpha\gamma}{\beta} = -\frac{q}{\beta^2}$$

$$(II) -\frac{b}{1} = \frac{\beta\gamma}{\alpha} + \frac{\alpha\gamma}{\beta} + \frac{\alpha\beta}{\gamma} = \frac{(\beta\gamma)^2 + (\alpha\gamma)^2 + (\alpha\beta)^2}{\alpha\beta\gamma}$$

$$\text{Now } [\beta\gamma + \alpha\gamma + \alpha\beta]^2 = \beta^2\gamma^2 + 2\alpha\beta\gamma^2 + 2\alpha\beta\gamma^2 + \alpha^2\gamma^2 + 2\alpha^2\beta\gamma + \alpha^2\beta^2 = \beta^2\gamma^2 + \alpha^2\gamma^2 + \alpha^2\beta^2 + 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$\text{Let } X = -\frac{q}{\alpha^2}$$

$$\therefore \alpha^2 = -\frac{q}{X}$$

$$\alpha = \pm \sqrt{\frac{-q}{X}}$$

$$\text{So } \beta^2\gamma^2 + \alpha^2\gamma^2 + \alpha^2\beta^2 = [\beta\gamma + \alpha\gamma + \alpha\beta]^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma) = (-3p)^2 - 2q \cdot 0 = 9p^2$$

$$\text{Gives } -b = \frac{9p^2}{-q} \quad \begin{array}{l} * \alpha + \beta + \gamma = 0 \\ * \alpha\beta + \alpha\gamma + \beta\gamma = -3 \\ * \alpha\beta\gamma = -q \end{array}$$

$$b = \frac{9p^2}{q}$$

On Substitution

$$\left[\pm \sqrt{\frac{-q}{X}}\right]^3 - 3p \left[\pm \sqrt{\frac{-q}{X}}\right] + q = 0$$

$$\pm \left(\frac{-q}{X}\right)^{\frac{3}{2}} - 3p \left(\pm \left(\frac{-q}{X}\right)^{\frac{1}{2}}\right) = -q$$

$$\begin{aligned} * \frac{c}{1} &= \frac{\beta\gamma}{\alpha} + \frac{\alpha\gamma}{\beta} + \frac{\alpha\beta}{\gamma} = \frac{\beta\gamma}{\alpha} + \frac{\alpha\gamma}{\beta} + \frac{\alpha\beta}{\gamma} \\ &= \frac{\beta^2\gamma^2}{\alpha\beta\gamma} + \frac{\alpha^2\gamma^2}{\alpha\beta\gamma} + \frac{\alpha^2\beta^2}{\alpha\beta\gamma} \\ &= \frac{(\beta\gamma)^2 + (\alpha\gamma)^2 + (\alpha\beta)^2}{\alpha\beta\gamma} \\ &= \frac{9p^2}{-q} = -\frac{9p^2}{q} \end{aligned}$$

Square both sides

$$\left(\frac{-q}{X}\right)^3 - 6p \left(\frac{-q}{X}\right)^{\frac{1}{2}} + q^2 \left(\frac{-q}{X}\right) = q^2$$

$$\therefore -\frac{q^3}{X^3} - 6pq^2X - q^2qX^2 = q^2X^3$$

$$\Rightarrow q^2X^3 + 6pq^2X^2 + 6pq^2X + q^3 = 0$$

$$\text{So } X^3 + \frac{6p}{q}X^2 + 6pX + q = 0$$

$$\text{Then } x^3 + bx^2 + cx + d = 0$$

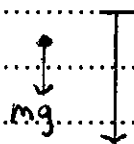
$$\Rightarrow x^3 + \frac{6p}{q}x^2 + 6px + q = 0$$

Start here for

Question Number:

15

(a) (i)



$$A = kv$$

at $t=0$

$$v=0$$

$$x=0$$

$$F = mg - kv$$

$$\text{So } m\ddot{x} = mg - kv$$

$$\ddot{x} = g - kv$$

Terminal velocity v_T is at $\ddot{x} = 0$

$$\text{So, let } g - kv_T = 0$$

$$v_T = \frac{g}{k}$$

$$\text{or } \frac{dv}{dt} = g - kv$$

$$\frac{dt}{dv} = \frac{1}{g - kv}$$

$$t = -\frac{1}{k} \int \frac{-k}{g - kv} dv$$

$$t = -\frac{1}{k} \left[\ln(g - kv) \right]_0^{v_T}$$

$$= -\frac{1}{k} \left[\ln(g - kv_T) - \ln(g) \right]$$

$$-kt = \ln \left[\frac{g - kv_T}{g} \right]$$

$$e^{-kt} = \frac{g - kv_T}{g}$$

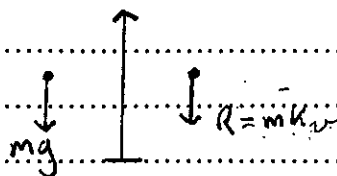
$$kv_T = g(1 - e^{-kt})$$

$$\text{as } t \rightarrow \infty, e^{-kt} \rightarrow 0$$

$$\therefore kv_T = g$$

$$v_T = \frac{g}{k}$$

(ii)



(α)

at $t=0$

$$x=0$$

$$v=v_T$$

$$m\ddot{x} = -mg - mkv$$

$$\ddot{x} = -g - kv$$

$$\text{Let } \frac{dv}{dt} = -g - kv$$

$$-\frac{dt}{dv} = \frac{1}{-g - kv}$$

$$t = -\frac{1}{k} \int_{v_T}^v \frac{-k}{-g - kv} dv$$

$$-kt = \left[\ln(-g - kv) \right]_{v_T}^v$$

$$-kt = \ln(-g - kv) - \ln(-g - kv_T)$$

$$= \ln \left[\frac{-g - kv}{-g - kv_T} \right]$$

$$e^{-kt} = \frac{g + kv}{g + kv_T}$$

$$\text{So } g e^{-kt} + e^{-kt} kv_T = g + kv$$

$$kv = g e^{-kt} + kv_T e^{-kt} - g$$

$$v = \frac{g}{k} e^{-kt} + v_T e^{-kt} - \frac{g}{k}$$

$$* v_T = \frac{g}{k}$$

$$= v_T e^{-kt} + v_T e^{-kt} - v_T$$

$$v = v_T (2e^{-kt} - 1)$$

$$(\beta) \text{ Let } v \frac{dv}{dx} = -(g + kv)$$

$$\frac{dv}{dx} = \frac{-(g + kv)}{v}$$

$$\frac{dx}{dv} = \frac{-v}{g + kv}$$

$$\begin{aligned}
 \text{Given } x &= \frac{1}{k} \int_{v_T}^0 \frac{k v}{g + k v} dv \\
 &= -\frac{1}{k} \int_{v_T}^0 \frac{g + k v - g}{g + k v} dv \\
 &= -\frac{1}{k} \left[\int_{v_T}^0 dv - \int_{v_T}^0 \frac{g}{g + k v} dv \right] \\
 &= -\frac{1}{k} \left[(v)_T^0 - \frac{g}{k} \int_{v_T}^0 \frac{k}{g + k v} dv \right] \\
 &= -\frac{1}{k} \left[(0 - v_T) - \frac{g}{k} [\ln(g + k v)]_{v_T}^0 \right] \\
 &= \frac{v_T}{k} + \frac{g}{k^2} [\ln(g + 0) - \ln(g + k v_T)] \\
 x &= \frac{v_T}{k} - \frac{g}{k^2} \ln \left[\frac{g + k v_T}{g} \right]
 \end{aligned}$$

$$\begin{aligned}
 * \quad v_T &= \frac{g}{k} \quad \text{So} \quad x = \frac{v_T}{k} \left[1 - \ln \left(1 + \frac{v_T}{v_T} \right) \right] \\
 &= \frac{v_T}{k} [1 - \ln 2]
 \end{aligned}$$

⑤ (i) Quadratic Formula gives

$$\begin{aligned}
 z &= \frac{\sin 2\theta \pm \sqrt{(\sin 2\theta)^2 - 4 \sin^2 \theta}}{2 \sin^2 \theta} \\
 &= \frac{2 \sin \theta \cos \theta \pm \sqrt{4 \sin^2 \theta \cos^2 \theta - 4 \sin^2 \theta}}{2 \sin^2 \theta} \\
 &= \frac{\cot \theta \pm \frac{2 \sin \theta \sqrt{\cos^2 \theta - 1}}{2 \sin^2 \theta}}{\sin \theta} \\
 &= \cot \theta \pm \frac{\sqrt{1 - \cos^2 \theta}}{\sin \theta}
 \end{aligned}$$

$$\therefore z = \cot \theta \pm \frac{i \sin \theta}{\sin \theta} \quad (1 - \cos^2 \theta = \sin^2 \theta)$$

$$= \cot \theta \pm i$$

$$\text{So } \alpha = \cot \theta - i \\ = \frac{\cos \theta - i \sin \theta}{\sin \theta}$$

$$\beta = \cot \theta + i \\ = \frac{\cos \theta + i \sin \theta}{\sin \theta}$$

$$\text{(ii.) } \alpha^n + \beta^n = \left[\frac{\cos \theta - i \sin \theta}{\sin \theta} \right]^n + \left[\frac{\cos \theta + i \sin \theta}{\sin \theta} \right]^n \\ = \frac{(\cos \theta - i \sin \theta)^n + (\cos \theta + i \sin \theta)^n}{\sin^n \theta}$$

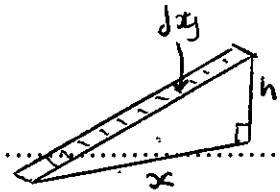
$$[\text{by de Moivre}] = \frac{\cos n\theta - i \sin n\theta + \cos n\theta + i \sin n\theta}{\sin^n \theta}$$

$$= \frac{2 \cos n\theta}{\sin^n \theta}$$

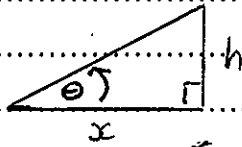
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Question Number: 16

(a)(i) Elemental cross-section slice



Area of face



$$\frac{h}{x} = \tan \theta$$

$$\text{AND } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$h = x \tan \theta \quad \text{--- (I)}$$

$$\therefore x^2 = \frac{a^2}{b^2} (b^2 - y^2) \quad \text{--- (II)}$$

$$\text{Now, Area} = \frac{1}{2} x h$$

$$= \frac{x^2 \tan \theta}{2} \quad \text{From (I) Area} = \frac{a^2}{b^2} (b^2 - y^2) \tan \theta$$

$$\therefore \text{Area} = \frac{a^2}{2b^2} (b^2 - y^2) \tan \theta$$

(ii) Elemental cross section has volume

$$\delta V = A \cdot \delta y$$

$$= \frac{a^2}{2b^2} (b^2 - y^2) \tan \theta \cdot dy$$

$$\text{So Volume} = \lim_{\delta y \rightarrow 0} \sum_{y=-b}^b \frac{a^2}{2b^2} (b^2 - y^2) \tan \theta \cdot dy$$

$$= \frac{a^2}{2b^2} \int_{-b}^b (b^2 - y^2) \tan \theta \, dy$$

(*tan θ constant)

[even fn]

$$= \frac{a^2 \tan \theta}{b^2} \int_0^b (b^2 - y^2) \, dy$$

$$\text{Volume} = \frac{a^2 \tan \theta}{b^2} \left[b^2 y - \frac{1}{3} y^3 \right]_0^b$$

$$= \frac{a^2 \tan \theta}{b^2} \left[b^3 - \frac{1}{3} b^3 \right] - 0$$

$$\text{Gives Volume as } \frac{2a^2 b \tan \theta}{3}$$

$$(b) (i) (c) \quad \frac{1}{x} - \frac{1}{x+1} = \frac{(x+1) - x}{x(x+1)}$$

$$= \frac{1}{x+1} \left[\frac{1!}{x+1} \right]$$

$$(f) \quad \frac{1}{x} - \frac{2}{x+1} + \frac{1}{x+2} = \frac{(x+1)(x+2) - 2x(x+2) + x(x+1)}{x(x+1)(x+2)}$$

$$= \frac{x^2 + 3x + 2 - 2x^2 - 4x + x^2 + x}{x(x+1)(x+2)}$$

$$= \frac{2}{x(x+1)(x+2)} \quad (2! = 2 \times 1 = 2)$$

$$(ii) \text{ L.H.S. } T(k, x) - T(k, x+1)$$

$$= \frac{k!}{x(x+1)(x+2)\dots(x+k)} - \frac{k!}{(x+1)(x+2)(x+3)\dots(x+k+1)}$$

$$= \frac{k!(x+k+1)}{x(x+1)(x+2)\dots(x+k)(x+k+1)}$$

$$= \frac{k!(k+1)}{x(x+1)(x+2)\dots(x+k+1)}$$

$$= \frac{(k+1)!}{x(x+1)(x+2)\dots(x+k+1)}$$

$$= T(k+1, x) \quad \text{R.H.S.}$$

$$(ii) \text{ For } n=1, \text{ L.H.S. } \frac{n c_0}{x} - \frac{n c_1}{x+1} = \frac{1}{x} - \frac{1}{x+1}$$

$$= \frac{1!}{x(x+1)} \quad \text{from (i)}$$

$$\text{R.H.S. is } \frac{1!}{x(x+1)} \quad \text{so true for } n=1$$

Let proposition be true for $n=k$ where k is a positive integer > 1

$$\text{Then } T(k, x) \Rightarrow \frac{k c_0}{x} - \frac{k c_1}{x+1} + \frac{k c_2}{x+2} + \dots + \frac{(-1)^k c_k}{x+k}$$

$$= \frac{k!}{x(x+1)(x+2)\dots(x+k)}$$

For next $n = k+1$, the L.H.S. is

$$\frac{{}^{k+1}C_0}{x} - \frac{{}^{k+1}C_1}{x+1} + \frac{{}^{k+1}C_2}{x+2} - \dots + (-1)^k \frac{{}^{k+1}C_k}{x+k} + (-1)^{k+1} \frac{{}^{k+1}C_{k+1}}{x+k+1}$$

Now ${}^{k+1}C_0 = {}^kC_0$ and ${}^{k+1}C_{k+1} = {}^kC_k$

Also, ${}^{k+1}C_r = {}^kC_r + {}^kC_{r-1}$

So L.H.S. becomes

$$\frac{{}^kC_0}{x} - \left[\frac{{}^kC_1 + {}^kC_0}{x+1} \right] + \left[\frac{{}^kC_2 + {}^kC_1}{x+2} \right] - \dots + (-1)^k \left[\frac{{}^kC_k + {}^kC_{k-1}}{x+k} \right] + (-1)^{k+1} \frac{{}^kC_k}{x+k+1}$$

$$= \frac{{}^kC_0}{x} - \frac{{}^kC_1}{x+1} + \frac{{}^kC_2}{x+2} - \dots + (-1)^k \frac{{}^kC_k}{x+k}$$

$$- \left[\frac{{}^kC_0}{x+1} - \frac{{}^kC_1}{x+2} + \frac{{}^kC_2}{x+3} - \dots + (-1)^k \frac{{}^kC_{k-1}}{x+k} + (-1)^{k+1} \frac{{}^kC_k}{x+k+1} \right]$$

$$= T(k, x) - T(k, x+1)$$

$$= T(k+1, x) \quad (\text{from (i)})$$

$$= \frac{(k+1)!}{x \cdot (x+1) \cdot (x+2) \cdot \dots \cdot (x+k+1)} \quad \text{as required.}$$

If true for $n=k$ it is true for $n=k+1$. So by method of induction true for all k .